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# Task 1: Data Cleaning & Preprocessing
# Dataset Used: Titanic Dataset

# Import Libraries
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

from sklearn.preprocessing import LabelEncoder, StandardScaler

# Load Dataset
# Download Titanic dataset from:
# https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv

df = pd.read_csv("/content/Titanic-Dataset.csv (1).xls")

# Display First 5 Rows
print("First 5 Rows:")
print(df.head())

# Basic Information
print("\nDataset Info:")
print(df.info())

# Check Missing Values
print("\nMissing Values:")
print(df.isnull().sum())

# -----
# Handling Missing Values
# -----

# Fill Age missing values with median
df['Age'].fillna(df['Age'].median(), inplace=True)

# Fill Embarked missing values with mode
df['Embarked'].fillna(df['Embarked'].mode()[0], inplace=True)

# Drop Cabin column because it has too many missing values
df.drop(columns=['Cabin'], inplace=True)

# Verify Missing Values Again
print("\nMissing Values After Cleaning:")
print(df.isnull().sum())

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# Convert Categorical Data to Numeric
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# Label Encoding for Sex column
le = LabelEncoder()
df['Sex'] = le.fit_transform(df['Sex'])

# One-Hot Encoding for Embarked column
df = pd.get_dummies(df, columns=['Embarked'], drop_first=True)

print("\nDataset After Encoding:")
print(df.head())

# -----
# Feature Scaling
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scaler = StandardScaler()

# Scale numerical columns
df[['Age', 'Fare']] = scaler.fit_transform(df[['Age', 'Fare']])

print("\nDataset After Scaling:")
print(df[['Age', 'Fare']].head())

# -----
# Outlier Detection Using Boxplots
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plt.figure(figsize=(12,5))

plt.subplot(1,2,1)
sns.boxplot(y=df['Age'])
plt.title("Boxplot of Age")

plt.subplot(1,2,2)
sns.boxplot(y=df['Fare'])
plt.title("Boxplot of Fare")

plt.tight_layout()
plt.show()

# -----
# Removing Outliers using IQR Method
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# Function to remove outliers
def remove_outliers(data, column):
    Q1 = data[column].quantile(0.25)
    Q3 = data[column].quantile(0.75)
    IQR = Q3 - Q1

    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR
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        return data[(data[column] >= lower) & (data[column] <= upper)]

# Remove Outliers from Fare
df = remove_outliers(df, 'Fare')

print("\nDataset Shape After Removing Outliers:")
print(df.shape)

# -----
# Final Cleaned Dataset
# -----

print("\nFinal Dataset:")
print(df.head())

# Save Cleaned Dataset
df.to_csv("cleaned_titanic.csv", index=False)

print("\nCleaned dataset saved as 'cleaned_titanic.csv'")
```

